

Business Insights Report (Exploratory Data Analysis)

Project Overview

This project performs Exploratory Data Analysis (EDA) on a marketing campaign dataset to understand customer demographics, purchasing behavior, and campaign effectiveness. The workflow includes data cleaning, visualization, statistical analysis, and business recommendations.

Data Cleaning

- Loaded the dataset using Pandas.
- Checked data types and dataset structure.
- Removed duplicate records.
- Filled missing values in the Income column using the median.
- Verified the cleaned dataset before analysis.

Key Findings

- Higher-income customers spend more across product categories.
- Wine products contribute the highest expenditure.
- Most customers did not respond positively to marketing campaigns.
- Married customers form one of the largest customer groups.
- Positive correlations exist among several product purchase categories.

Recommendations

- Focus premium campaigns on high-income customers.
- Personalize marketing offers based on purchase history.
- Improve campaign targeting to increase response rates.
- Use customer segmentation to optimize marketing budget allocation.

Conclusion

The EDA provides actionable insights into customer behavior and purchasing trends that can support more effective marketing strategies and data-driven business decisions.